Peichun Hua

Computer Science and Engineering, School of Data Science
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EDUCATION

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) — September 2022 — June 2026
B.Eng in Computer Science and Engineering, School of Data Science (SDS)
First Class Honors, Outstanding Graduate
Overall GPA: 3.87/4.00 — Major GPA: 4.00/4.00 — Rank: 2/132

University of North Carolina, Chapel Hill — September 2024 - May 2025
Overall GPA: 4.00/4.00
Exchange Student — Computer Science Department

RESEARCH INTERESTS

- Responsible AI; Adversarial Machine Learning; (ACL 2026, Work Under Review)
- AI and IoT Security; Secure Hardware Design; (ACSAC 2025, ISCAS 2026, SMC 2026 Under Review, TODAES Under Review)
- Privacy-Enhancing Technologies: Differential Privacy (DP), Trusted Execution Environment (TEE), and Secure Multi-Party Computation (MPC); (Work Under Review)

PUBLICATION

Wentao Du, Siyu Wang, **Peichun Hua**, Linden Chen, Ram Sundara Ramen, Yunming Xiao, Diwen Xue. "The Limits of Passive Network-Stack Fingerprinting in Web Fraud Detection." *Under Review.*

Peichun Hua, Yunming Xiao. "PENUMBRA: Representation-Driven Scaling for Private Outsourced Retrieval." *Under Review.*

Junguang Yao, **Peichun Hua**, Yue Zheng. "Enroll Once, Identify Everywhere: Few-Shot Cross-Scenario Visual Speaker Recognition via Event-driven SlowFast Networks" *Under Review.*

Peichun Hua, Danyang Chen, Diwen Xue, Mingyu Li, Yunming Xiao. "Rediscovering Deep Hashing in Secure Outsourced Dense Retrieval." *Under Review.*

Peichun Hua, Yue Zheng. "ChaosFormer: Hardware-Bound Edge Transformer Model Obfuscation for Intellectual Property Protection." *Under Review, ACM Transactions on Design Automation of Electronic Systems (TODAES)*, Paper.

Danyang Chen, Yufeng Gu, Yibo Huang, Chengxuan Pei, **Peichun Hua**, Yang Zhou, Yunming Xiao. "Unlocking Software-defined GPU Fabric Scheduling in the LLM Era." *Under Review.*

Yanchuan Tang, Junnan Li, Zhan Cheng, **Peichun Hua**, Skylar Zhai, TianMing Sha, Yuan Gao, Ruixiang Tang, Yuanyuan Lei. "MAGMA: Defending Federated Learning via Intrinsic Manifold Geometry." *Under Review.*

Xin Wang, **Peichun Hua**, Chip Hong Chang, Wenye Liu, Yue Zheng. "A Unified Open-Set Framework for Scalable PUF-Based Authentication of Heterogeneous IoT Devices." To appear at the IEEE International Conference on Systems, Man, and Cybernetics (SMC 2026) (CORE: B, Paper (arXiv)).

Peichun Hua, Hao Li, Shanghao Shi, Zhiyuan Yu, Ning Zhang. "Rethinking Jailbreak Detection of Large Vision Language Models with Representational Contrastive Scoring." To appear at the Annual Meeting of the Association for Computational Linguistics (ACL 2026). (**Main Conference, Oral Presentation**, CORE: A*, AR = ∼2300/12148 ≈ 19%, Code, Paper).

Peichun Hua, Hanxiu Zhang, Tuo Li, Yue Zheng, Wenye Liu. "Live Demonstration: Hardware-Bound IP Protection for Edge-deployed Transformers." To appear at the IEEE International Symposium on Circuits and Systems (ISCAS 2026) (CORE: C, AR = 1032/2012 ≈ 51.3%, Paper).

Peichun Hua, Hanxiu Zhang, Tuo Li, Yue Zheng. "Securing On-device Transformer with Hardware Binding and Reversible Obfuscation." In the Annual Computer Security Applications Conference (ACSAC 2025) (CORE: A, AR = 84/446 ≈ 18.8%, Code, Paper, Presentation).

AWARDS

Academic Honors

- **Outstanding Graduate**, School of Data Science, CUHK-Shenzhen — 2026
- Dean's List, School of Data Science, CUHK-Shenzhen — 2022–2023, 2023–2024
- Academic Performance Scholarship (Class B, Top <2%), School of Data Science — 2023–2024
- Academic Performance Scholarship (Class C, Top <5%), School of Data Science — 2022–2023
- University Research Award (with scholarship) — Fall 2024, Spring 2025

Research Awards

- *Student Conferenceship*, 41st Annual Computer Security Applications Conference, ACSAC 2025

- *Artifact Available*, *Artifact Reviewed*, and *Artifact Reproducible* Badges — IEEE, for the paper "Securing On-device Transformer with Hardware Binding and Reversible Obfuscation" at ACSAC 2025
- International Student Research Internship Program (Fully Funded), McKelvey School of Engineering, Washington University in St. Louis — Summer 2025

EXPERIENCE

University of Southern California — Los Angeles, CA, USA
Research Assistant — May 2026 – Present

- Advisor: Prof. Mengyuan Li
- Topic: Agentic AI Security.

School of Data Science, CUHK-Shenzhen — Shenzhen, China
Research Assistant — September 2024 – Present

- Advisor: Prof. Yunming Xiao
- Topic: Utilizing Differential Privacy, Trusted Execution Environment, and Deep Hashing techniques to create secure outsourced dense retrieval in the RAG pipeline, with strong defense against privacy attacks while retaining performance and high throughput.
- Work completed independently. Resulting paper is under review.

McKelvey School of Engineering, Washington University in St. Louis — Missouri, MO, USA
Summer Research Intern — May 2025 – August 2025

- Fully funded by International Student Research Internship Program
- Advisor: Prof. Ning Zhang
- Topic: Multi-modal Jailbreak Detection with interpretability methods instantiated with statistical tests and contrastive learning. Achieve highly effective, generalizable, and lightweight attack detection.
- Work completed independently. **Paper accepted by ACL 2026 Oral.**

School of Science and Engineering, CUHK-Shenzhen — Shenzhen, China
Undergraduate Research Intern — May 2024 – May 2025

- Advisor: Prof. Yue Zheng
- Topic: Protection of the IP (intellectual property) of on-device transformer models with a specialized encryption scheme and a hardware-software co-design approach to effectively prevent unauthorized usage at the edge.
- Conferred University Research Award (URA, with scholarship). Work completed independently and directly accepted by ACSAC 2025. Live Demonstration paper accepted by IEEE ISCAS 2026; the extended journal version under review at ACM TODAES.

SKILLS

- **Coursework:** Achieved A grade in all listed *graduate-level* coursework

Cryptography	3D Computer Vision	Hardware Security and Side-Channels
Efficient Deep Learning	Research Topics in Computer Security	Formal Methods in Computer Security

- **Language Proficiency:** TOEFL 108 (R29+L29+S23+W27), GRE 328 (V161+Q167+3.5)

SERVICES

- Reviewer, IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)
- Reviewer, ACL Rolling Review